



Customer Purchase Behavior Analysis

Uncovering insights from 3,900 transactions to guide strategic business decisions

Dataset Overview

3,900

Transactions

Purchase records
analyzed

18

Features

Data points per
transaction

37

Missing Values

Review ratings imputed

Customer demographics, purchase details, and shopping behavior analyzed



Python Data Preparation

Data Loading

Imported using pandas, checked structure with info() and describe()

Standardization

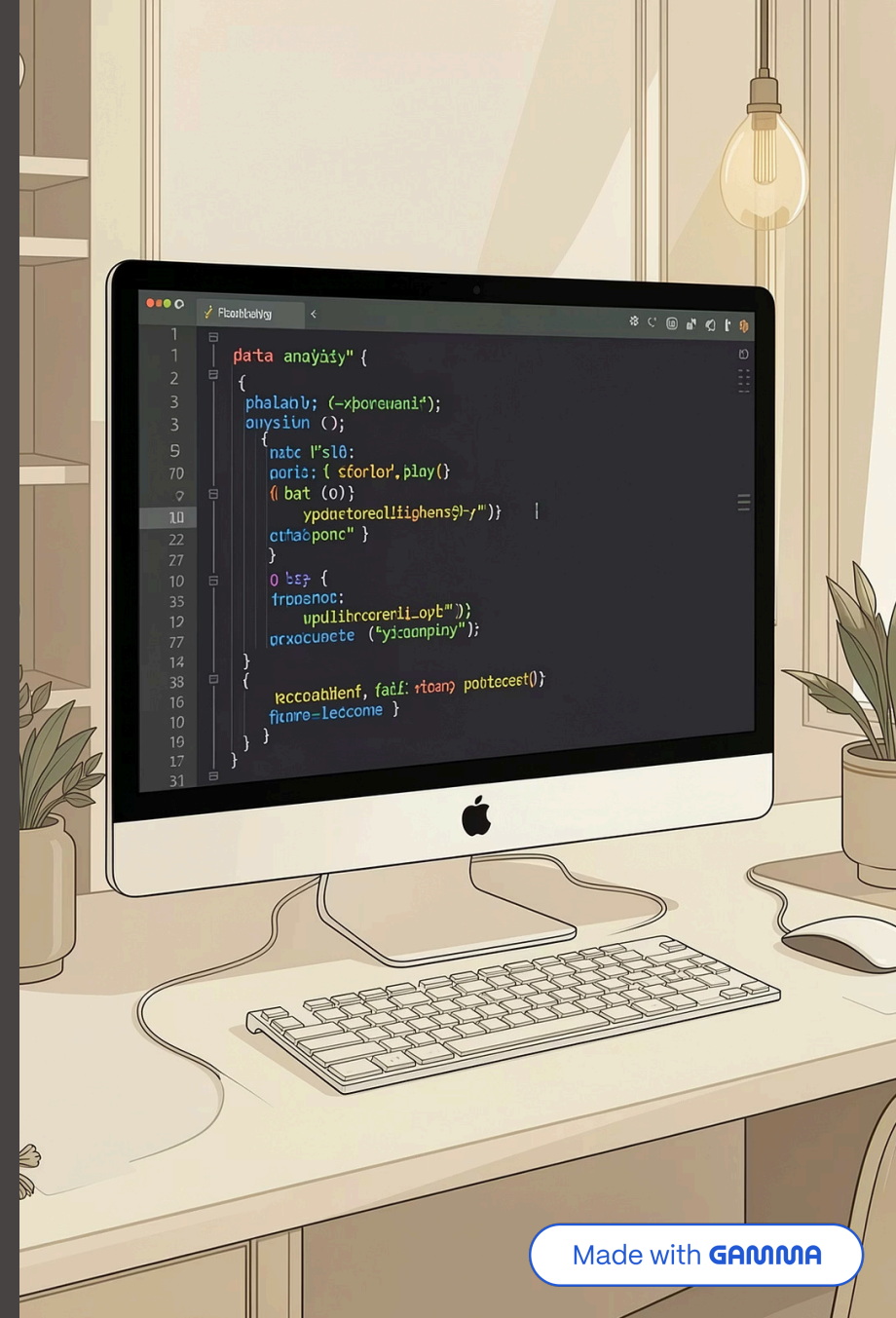
Renamed columns to snake case for readability

Cleaning

Imputed missing review ratings using median by category

Feature Engineering

Created age groups and purchase frequency calculations



SQL Analysis: Key Findings



Revenue by Gender

Male: \$157,890 | Female: \$75,191



Top Products

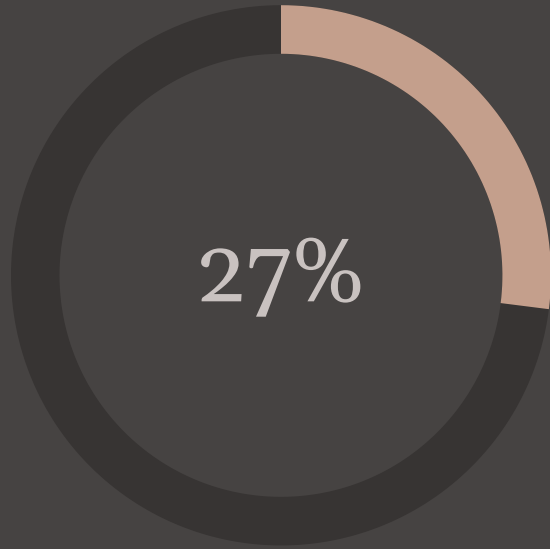
Gloves (3.86), Sandals (3.84), Boots (3.82)



Shipping Impact

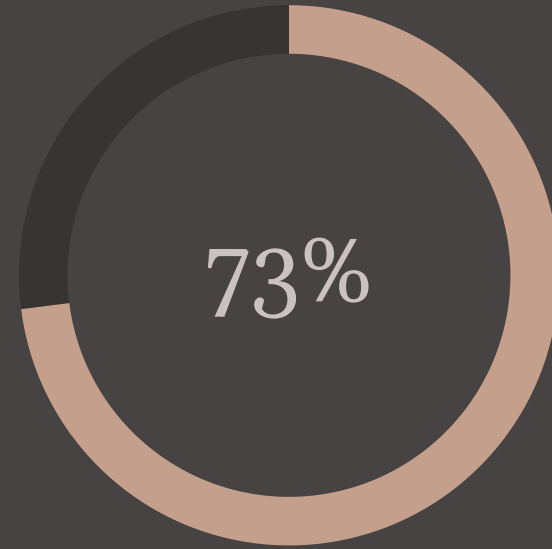
Express: \$60.48 | Standard: \$58.46

Subscription Insights



Subscribers

1,053 customers



Non-Subscribers

2,847 customers

Non-subscribers generate higher total revenue (\$170,436 vs \$62,645)

Discount Strategy Analysis

1

High-Spending Discount Users

839 customers used discounts while spending above average

2

Discount-Dependent Products

Hat (50%), Sneakers (49.66%), Coat (49.07%) most discounted

3

Revenue Impact

Balance sales boosts with margin control

Customer Segmentation



New Customers

83 customers



Returning

701 customers



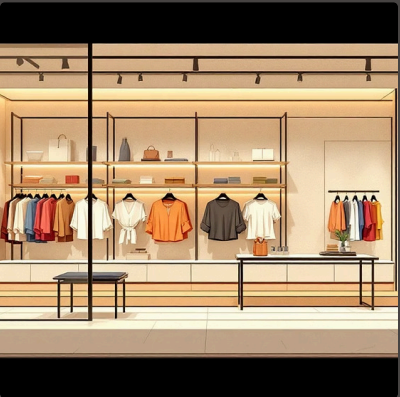
Loyal

3,116 customers

Repeat buyers with >5 purchases more likely to subscribe

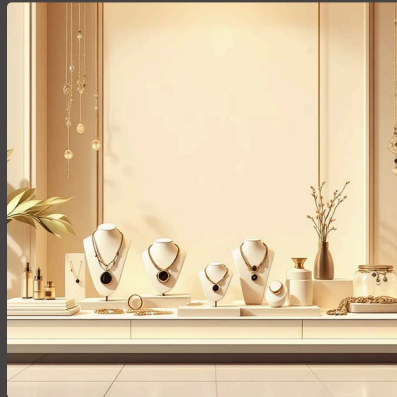


Product Performance



Blouse

171 orders - Top clothing item



Jewelry

171 orders - Top accessory



Belt

161 orders - Tied for accessories

Age Group Revenue



Young Adult

\$62,143 revenue



Middle-Aged

\$59,197 revenue



Adult

\$55,978 revenue



Senior

\$55,763 revenue

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits



Loyalty Programs

Reward repeat buyers



Review Discount Policy

Balance margins



Targeted Marketing

Focus on high-revenue groups

